ELK 346 E
Data Acquisition & Embedded Systems

2023 – 2024 Spring Semester

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Topics

- Sensors
- Filters, ADCs, DACs
- Sampling
- Fundamental Digital Electronics
- Basics of Microcontrollers – Registers, ALU
- ARM - Introduction
- ARM – Registers
- Embedded Programming

Filters

- Separation of a wanted component of a signal from the unwanted one. (i.e. telecommunication signal applications)
- Passive filters (R, L, C)
- Active filters (Op-Amps)

$$H(\omega) = \frac{output\ voltage}{input\ voltage}$$

Passive / Active Filters

Passive Filters

Advantages:

- Do not require a power supply to operate.
- Have a simple design and are relatively easy to implement.
- Do not introduce noise or distortion into the signal.
- Are generally less expensive than active filters.

Disadvantages:

- Have limited filtering capabilities, particularly in the high-frequency range.
- Can cause loading effects on the signal source due to their high output impedance.
- Cannot provide amplification as well as filtering.

Active Filters

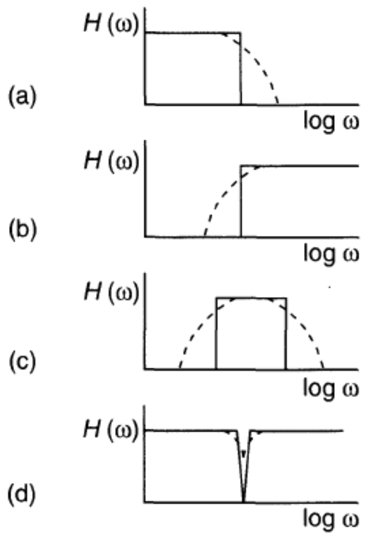
Advantages:

- Can provide amplification as well as filtering, allowing for greater flexibility in signal processing.
- Have low output impedance, which makes them less susceptible to loading effects.
- Can be designed with sharp cutoff characteristics.
- Can be easily tuned to specific frequency ranges.

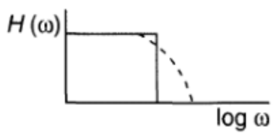
Disadvantages:

- Require a power supply to operate.
- Can introduce noise and distortion into the signal.
- Can be more complex and expensive to design and implement.

Filter Responses

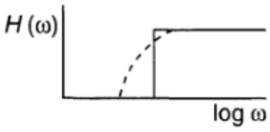


- a) Low-pass filter
- b) High-pass filter
- c) Band-pass filter
- d) Band-stop (Notch) filter



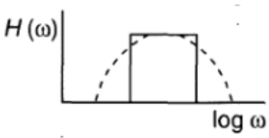
Low-Pass Filters

- Low pass filters attenuate higher frequencies.
- Audio systems: to remove high-frequency noise from signals. This can help improve the clarity and quality of the audio signal.
- Power supplies: to reduce high-frequency noise that may be introduced into the output voltage.
- Signal processing: to remove high-frequency noise or unwanted signal components.
- Antenna design: to remove high-frequency noise from signals received by the antenna.
- Image processing: to smooth out noise or reduce the level of high-frequency detail in an image.
- Data communications: to remove high-frequency noise from signals that may interfere with the transmission of data.
- Motor control: to filter out high-frequency noise that may be introduced into the motor control signals.



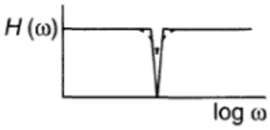
High-Pass Filters

- High pass filters attenuate lower frequencies.
- Audio systems: to remove low-frequency noise or unwanted signal components from signals. This can help improve the clarity and quality of the audio signal.
- Signal processing: to remove low-frequency noise or unwanted signal components.
- Antenna design: to remove low-frequency noise from signals received by the antenna.
- Image processing: to enhance the level of high-frequency detail in an image.
- Data communications: to remove low-frequency noise from signals that may interfere with the transmission of data.
- Motor control: to filter out low-frequency noise that may be introduced into the motor control signals.
- Vibrational analysis: to remove low-frequency noise from signals obtained from sensors that detect mechanical vibrations.



Band-Pass Filters

- Band pass filters allow signals within a specific frequency range to pass through while attenuating signals outside of that range.
- Radio frequency (RF) applications: commonly used in RF applications such as radio receivers to filter out unwanted signals outside of the desired frequency range.
- Wireless communications: used in wireless communications systems to isolate specific frequency bands, such as the frequency bands used for Wi-Fi, Bluetooth, and cellular communications.
- Biomedical applications: used in biomedical applications such as electroencephalography (EEG) and electrocardiography (ECG) to filter out unwanted noise and interference from the signal.
- Audio systems: used in audio systems to isolate specific frequency bands, such as bass or treble, and remove unwanted signals outside of that band.
- Instrumentation: used in instrumentation applications to isolate specific frequency bands in signals from sensors, such as accelerometers or strain gauges.
- Seismic analysis: used in seismic analysis applications to isolate specific frequency bands in signals from seismometers that detect ground vibrations.
- Industrial process control: used in industrial process control applications to filter out unwanted signals outside of a specific frequency range in signals from sensors used to monitor the process.



Band-Stop Filters

- Band stop or Notch filters attenuate signals within a specific frequency range while allowing signals outside of that range to pass through.
- Radio frequency (RF) applications: commonly used in RF applications to eliminate unwanted signals or interference within a specific frequency range, such as those caused by nearby transmitters.
- Audio systems: used in audio systems to eliminate unwanted signals or interference within a specific frequency range, such as those caused by 60Hz electrical hum.
- Biomedical applications: used in biomedical applications to eliminate unwanted signals or interference within a specific frequency range, such as those caused by power line noise or muscle artifacts in EEG signals.
- Instrumentation: used in instrumentation applications to eliminate unwanted signals or interference within a specific frequency range, such as those caused by electromagnetic interference (EMI) or radio frequency interference (RFI).
- Speech processing: used in speech processing applications to eliminate unwanted signals or interference within a specific frequency range, such as those caused by background noise or cross-talk between microphones.
- Audio feedback suppression: used in audio feedback suppression systems to eliminate unwanted signals within a specific frequency range, such as those caused by acoustic feedback between a microphone and a loudspeaker.
- Power electronics: used in power electronics applications to eliminate unwanted signals or interference within a specific frequency range, such as those caused by switching transients or harmonic distortion.

Active Filters

- All the passive filter applications can be implemented as active filters using op-amps.
- Do research:
  - Active high-pass filter
  - Active low-pass filter
  - Active band-pass filter
  - Active band-stop filter

Digital Filters

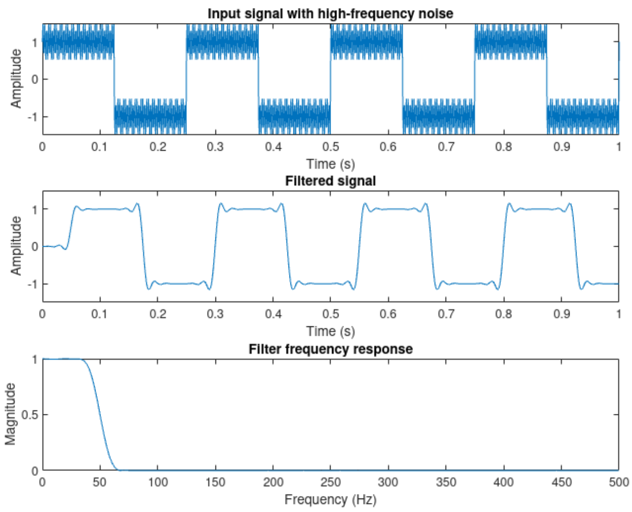
- Finite Impulse Response (FIR) filter: This type of digital filter has a finite number of coefficients and uses only the input signal to calculate the output signal. FIR filters are easy to design, have linear phase response, and are often used in audio processing.

- Infinite Impulse Response (IIR) filter: This type of digital filter uses both the input signal and the previous output signal to calculate the current output signal. IIR filters can have nonlinear phase response, but are computationally efficient and are often used in control systems and communication systems.

FIR Filter Example

```
% Design a low-pass FIR filter
fs = 1000; % Sampling frequency in Hz
fc = 50; % Cutoff frequency in Hz
numtaps = 100; % Number of filter taps
b = fir1(numtaps, fc/(fs/2)); % Compute
the filter coefficients using the FIR1
function

% Generate a square wave input signal
with a frequency of 4 Hz
t = 0:1/fs:1;
x = square(2*pi*4*t);

% Add high-frequency noise to the input
signal
noise = 0.5*sin(2*pi*300*t); % High-
frequency noise with a frequency of 300
Hz
x = x + noise;

% Filter the input signal using the FIR
filter
y = filter(b, 1, x);
```

```
% Plot the results
figure;
subplot(3,1,1);
plot(t,x);
title('Input signal with high-frequency noise');
xlabel('Time (s)');
ylabel('Amplitude');
ylim([-1.5, 1.5]);
subplot(3,1,2);
plot(t,y);
title('Filtered signal');
xlabel('Time (s)');
ylabel('Amplitude');
ylim([-1.5, 1.5]);

% Plot the frequency response of the filter
[H,f] = freqz(b,1,1024,fs);
subplot(3,1,3);
plot(f,abs(H));
title('Filter frequency response');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
```



IIR Filter Example

```
% Design a second-order Butterworth low-
pass IIR filter
fs = 1000; % Sampling frequency in Hz
fc = 50; % Cutoff frequency in Hz
[b,a] = butter(2, fc/(fs/2)); % Compute the filter
coefficients using the BUTTER function

% Generate a square wave input signal with a
frequency of 4 Hz
t = 0:1/fs:1;
x = square(2*pi*4*t);

% Add high-frequency noise to the input signal
noise = 0.5*sin(2*pi*300*t); % High-frequency
noise with a frequency of 300 Hz
x = x + noise;

% Filter the input signal using the IIR filter
y = filter(b, a, x);
```

```
% Plot the results
figure;
subplot(3,1,1);
plot(t,x);
title('Input signal with high-frequency noise');
xlabel('Time (s)');
ylabel('Amplitude');
ylim([-1.5, 1.5]);
subplot(3,1,2);
plot(t,y);
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```
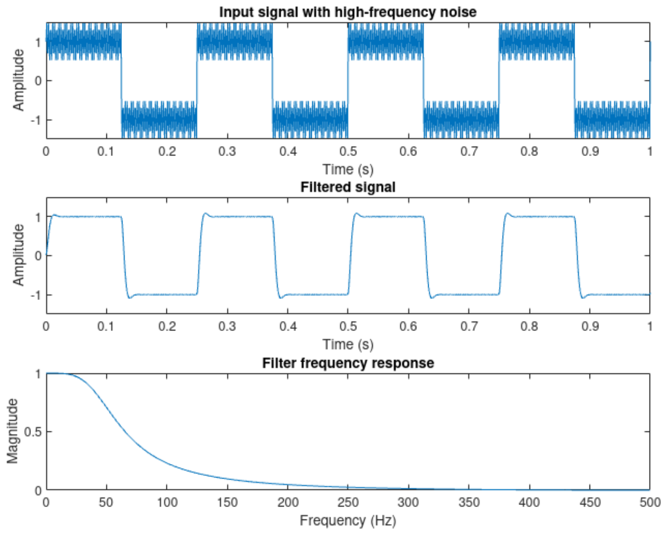



Analog Digital Converters

- An Analog-to-Digital Converter (ADC) is an electronic component or system that converts an analog signal (continuous electrical signal) into a digital signal (discrete numerical representation).
- The process of analog-to-digital conversion involves sampling, quantization, and encoding.
  - First, the analog signal is sampled at regular intervals,
  - Then the amplitude of each sample is quantized into a discrete value.
  - Finally, the quantized values are encoded into binary digits (bits) that can be stored, transmitted, or processed by a digital system.

ADC Types

- Successive Approximation ADC (SAR ADC): This is a type of ADC that uses a binary search algorithm to approximate the input signal. It is simple, fast, and widely used in microcontroller-based systems.
- Delta-Sigma ADC: This is a type of ADC that uses oversampling and noise shaping techniques to achieve high resolution and accuracy. It is commonly used in audio applications, such as digital audio recording and playback.
- Ramp ADC: This is a type of ADC that uses a voltage ramp to measure the input signal. It is relatively simple and fast but has limited resolution and accuracy.
- Flash ADC: This is a type of ADC that uses a bank of comparators to compare the input signal with a set of reference voltages. It is fast and accurate but requires a large number of comparators and consumes high power.
- Pipeline ADC: This is a type of ADC that uses a series of stages to divide the conversion process into smaller and faster sub-conversions. It is fast and accurate but requires more complex circuitry and has higher latency.
- Time-Interleaved ADC: This is a type of ADC that uses multiple channels and interleaving techniques to increase the sampling rate and bandwidth. It is commonly used in high-speed data acquisition systems.

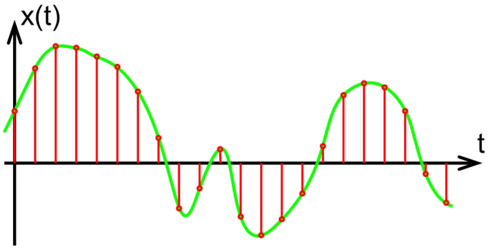
Digital Analog Converters

- A Digital-to-Analog Converter (DAC) is an electronic component or system that converts a digital signal (discrete numerical representation) into an analog signal (continuous electrical signal).
- The process of digital-to-analog conversion involves decoding, weighting, and summation.
  - First, the digital signal is decoded into a set of binary weights that represent the amplitude of each analog component.
  - Then, the binary weights are multiplied by corresponding analog components
  - Finally, summed to generate the final analog signal.

Analog Digital Converters

Sampling:

- In the sampling step, the continuous analog signal is measured at regular intervals. This process involves capturing the value of the analog signal at specific points in time. The following graph illustrates the sampling process:

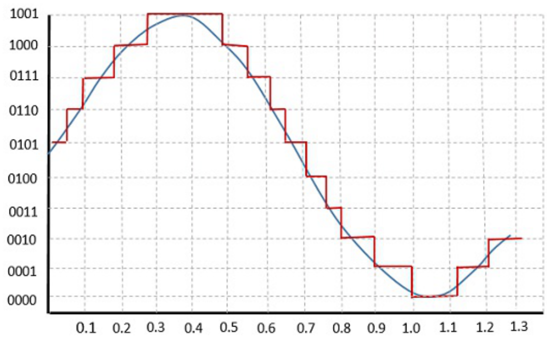


- In this graph, the continuous analog signal (green curve) is sampled at regular intervals. Each red dot represents a discrete sample of the analog signal.

Analog Digital Converters

Quantization:

- After sampling, the analog values are quantized into discrete digital values. The number of bits used for quantization determines the resolution of the ADC. A higher bit count allows for finer quantization and better representation of the analog signal. Here's a graphical representation:

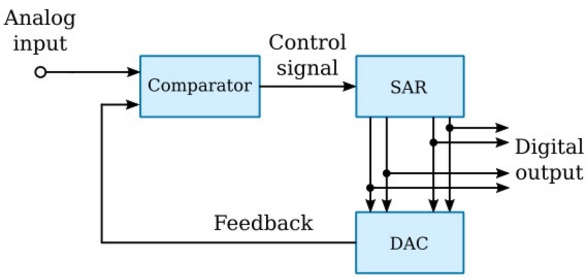


- In this example, the analog signal is quantized into 4-bit digital values. The horizontal lines represent the quantization levels, and each sample is assigned to the nearest quantization level.

Analog Digital Converters

Conversion:

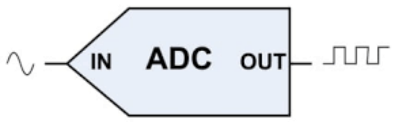
- The quantized digital values are then converted into binary format. One common method used for this conversion is successive approximation. The following graph shows the successive approximation process:
- In each step of the successive approximation, the digital value is refined until a close match to the analog signal is achieved. The process continues until the desired level of accuracy is reached.



Analog Digital Converters

Output:

- Finally, the resulting digital value is available for the microcontroller to use. This digital value can be processed further, stored, or used for control decisions. The output is a representation of the original analog signal in a digital form.



Digital Analog Converters

- A DAC (Digital-to-Analog Converter) in a microcontroller performs the opposite function of an ADC
- It converts digital signals into analog signals.
- This is important in applications where the microcontroller needs to interface with devices that operate in the analog domain, such as audio devices, motor controllers, or analog sensors.

Digital Analog Converters

- Here's a breakdown of how a DAC works in a microcontroller:
- Digital Input: The DAC receives a digital input from the microcontroller. This digital input typically consists of binary values, where each bit represents a different power of two. For example, an 8-bit DAC has 256 possible digital values (2^8).
- Conversion: The digital input is converted into an analog voltage or current. The DAC uses its internal circuitry to create a continuous analog signal that corresponds to the digital input. This process is often achieved through resistor networks, current sources, or other electronic components.
- Output: The resulting analog signal is available at the DAC output. This analog signal can then be used to drive external devices that require analog input.
- Voltage Range and Resolution: DACs have a specified voltage range, often determined by a reference voltage. The digital input values are mapped to this voltage range. The resolution of the DAC, measured in bits, determines the granularity of the output voltage levels.
- Update Rate: DACs have an update rate or settling time, indicating how quickly they can respond to changes in the digital input. This is essential in applications where the analog output needs to follow changes in the digital input quickly.

DAC Types

- Voltage DAC: This type of DAC generates a voltage output that is proportional to the digital input code.
- Current DAC: This type of DAC generates a current output that is proportional to the digital input code.
- R-2R Ladder DAC: This type of DAC uses a network of resistors arranged in a binary-weighted configuration to generate the analog output.
- Sigma-Delta DAC: This type of DAC uses oversampling and noise-shaping techniques to achieve high resolution and accuracy.
- PWM DAC: This type of DAC generates an analog output by modulating the pulse width of a high-frequency digital signal.

An ADC Example Scenario

Temperature Sensor → ADC → MCU → Control Algorithm → Heating System

- One example application of a temperature sensor read by an ADC and a microcontroller in a thermostat for a heating or cooling system. In this application, the temperature sensor measures the temperature of the environment, and the microcontroller uses the ADC to convert the analog voltage signal from the temperature sensor into a digital signal that can be processed by the microcontroller.
- The temperature sensor measures the temperature of the environment and generates an analog voltage signal proportional to the temperature. The ADC converts the analog voltage signal into a digital signal that can be read and processed by the microcontroller.
- The microcontroller uses a control algorithm to determine whether to turn on or off the heating or cooling system, based on the measured temperature and the desired temperature setpoint. The microcontroller can also perform other functions, such as displaying the temperature on a screen or sending alerts if the temperature exceeds a certain threshold.
- The heating or cooling system responds to the control signal from the microcontroller, either by turning on or off the heating or cooling elements, or by adjusting the flow of air or water to maintain the desired temperature.
- Overall, this system provides an automated way to control the temperature of a room or building, based on the input from a temperature sensor and the processing power of a microcontroller.

An ADC Example Scenario

Temperature Sensor → ADC → MCU → Control Algorithm → Heating System

- Temperature sensor that could be used in this application is the TMP36, which is a low-voltage temperature sensor that outputs an analog voltage proportional to the temperature in Celsius. The TMP36 is specified from −40°C to +125°C, provides a 750 mV output at 25°C,
- For the ADC, a common choice is the MCP3008, which is an 8-channel, 10-bit ADC with a SPI interface. The MCP3008 has a resolution of 10 bits, which means it can represent 1024 different voltage levels, and a conversion rate of up to 200 kSPS (kilosamples per second).
- Both the TMP36 and MCP3008 are widely available and relatively inexpensive, making them a good choice for many applications. However, it's important to note that there are many other temperature sensors and ADCs available on the market, and the specific choice will depend on the requirements of the application.

Check the datasheets of the components given above and elaborate the scenario by yourself.

An ADC/DAC Example Scenario

Mic → Analog Circuitry → ADC → DSP → DAC → Amplifier → Speaker

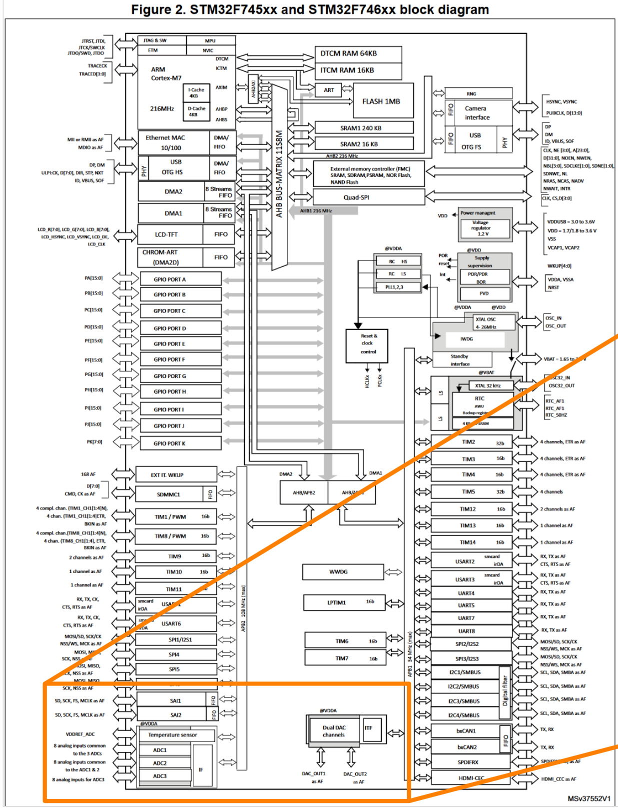
- Consider a musician recording a guitar track. The sound waves from the microphone are converted into an electrical signal, which is processed by the analog circuitry to amplify and filter the signal.
- The signal is then sampled by the ADC, which converts the continuous analog signal into a discrete digital signal that can be processed by the digital signal processing unit.
- The digital signal processing unit can perform various operations on the digital signal, such as filtering, equalization, and compression.
- Once the signal processing is complete, the digital signal is sent to the DAC, which converts it back into an analog signal that can be amplified and sent to the speakers or headphones.
- The amplifier boosts the power of the analog signal to a level that can drive the speakers or headphones, producing the sound waves that can be heard by the musician.

A Practical Example

Audio EQ Software Implementation (STM32)

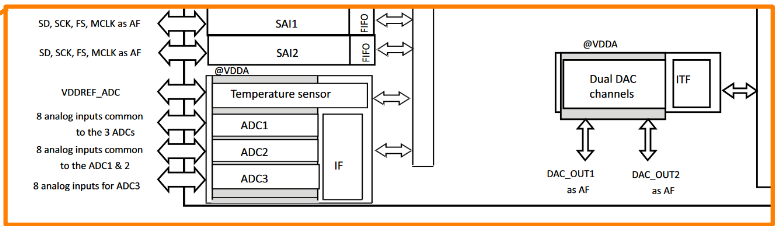
https://www.youtube.com/watch?v=4o-_gUht_Xc

Figure 2. STM32F745xx and STM32F746xx block diagram



STM32F746X

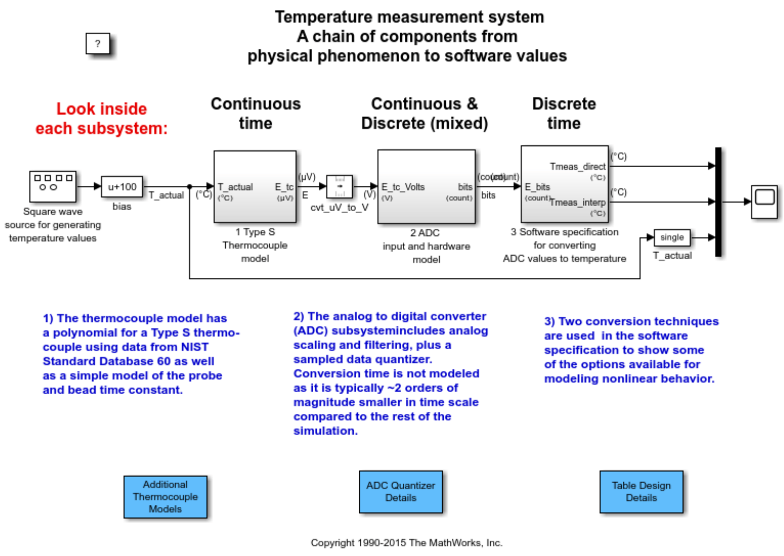
Check the datasheet for your own MCU!



Research

- Do research about the topics below for ADCs and DACs.
  - Integral Nonlinearity
  - Differential Nonlinearity
  - Resolution
  - Conversion Time

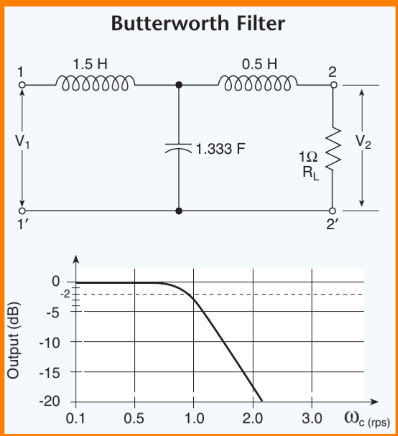
Check by yourself



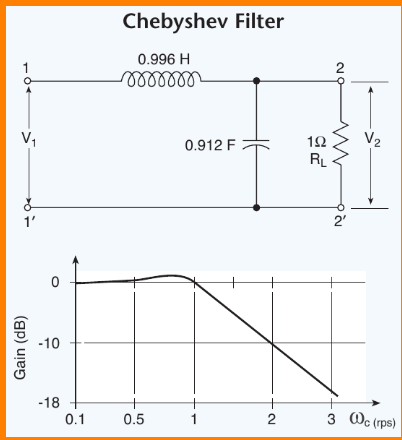
- Run in MATLAB
  - open_system('sldemo_tc')

https://www.mathworks.com/help/simulink/slref/approximating-nonlinear-relationships-type-s-thermocouple.html

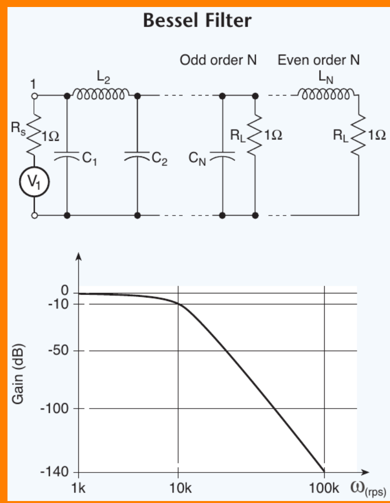
Common Passive Filter Types



This is a type of low-pass or high-pass filter with a maximally flat response in the passband. It is commonly used in audio and video applications.



This is a type of filter that provides a sharper cutoff than a Butterworth filter, but has ripples in the passband or stopband. It is commonly used in applications where a sharp cutoff is required.



This is a type of low-pass filter with a maximally linear phase response in the passband. It is commonly used in audio applications where phase distortion must be minimized